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0.1 Previous implementation

Definition 0.1. (Membership relation class)

$$\mathrm{E} = \{(x, y) | x \in y\}$$

Theorem 0.1.

$$\mathrm{E}^{-1}[A] = \bigcup A$$

(or even "$\exists [A] = \bigcup A$")

Proof.
($\subseteq$)
Assume $w \in \mathrm{E}^{-1}[A]$.
Therefore $\exists a \in A : (a, w) \in \mathrm{E}^{-1}$.
So $(w, a) \in \mathrm{E}$.
So $w \in a$.
Since $w \in a$ and $a \in A$, $w \in \bigcup A$.
($\supseteq$)
$w \in \bigcup A$.
Therefore $\exists a \in A : w \in a$.
So $(w, a) \in \mathrm{E}$. So $(a, w) \in \mathrm{E}^{-1}$. From $(a, w) \in \mathrm{E}^{-1}$ and $(a \in A)$, we have $(w \in \mathrm{E}^{-1}[A])$.
Done.

Definition 0.2. (Image)

$$F[A] := \{b | \exists a \in A : (a, b) \in F\}$$

Definition 0.3. (?)

$$F\langle A \rangle = \{b | \forall a \in A : (a, b) \in F\}$$

Theorem 0.2.

$$\mathrm{E}^{-1}\langle A \rangle = \bigcap A$$

Proof.

$$\begin{aligned} v \in \mathrm{E}^{-1}\langle A \rangle &\iff \forall a \in A : (a, v) \in \mathrm{E}^{-1} \\ &\iff \forall a \in A : (v, a) \in \mathrm{E} \\ &\iff \forall a \in A : v \in a \\ &\iff v \in \bigcap A \end{aligned}$$

Done.

Theorem 0.3. ()
$(\mathrm{dom}(\Sigma F)) \cup (F^{-1}[\{\emptyset\}]) = \mathrm{dom}(F)$
$(\mathrm{dom}(\Sigma F)) \cap (F^{-1}[\{\emptyset\}]) = \emptyset$

Proof.
Done.

Theorem 0.4. (Dunaev)
If $(F \subseteq V \times V)$ and (F functional) and $(\emptyset \notin \mathrm{ran}(F))$ then

$$F = \{(x, (\Sigma F)[\{x\}]) | x \in \mathrm{dom}(\Sigma F)\}$$

Proof.
Done.

Theorem 0.5.

$$(F \circ G)[X] = F[G[X]]$$

Proof.
Done.